STAT GU4204 — Statistical Inference

Final Examination (Version D) — *Solutions*

Hypothesis-Testing Centered

Spring 2026

Part A — Conceptual on Testing (10 points)

A1. (3 pts) Definitions and relations.

Let δ have critical region S_1 (and acceptance region S_0).

(i) **Type I error:** the event of rejecting H_0 when H_0 is true. Its probability is $\mathbb{P}_\theta(\mathbf{X} \in S_1)$ for $\theta \in \Omega_0$.

(ii) **Type II error:** the event of failing to reject H_0 when H_1 is true. Its probability is $\mathbb{P}_\theta(\mathbf{X} \in S_0) = 1 - \mathbb{P}_\theta(\mathbf{X} \in S_1)$ for $\theta \in \Omega_1$.

(iii) **Power function:** $\pi(\theta \mid \delta) = \mathbb{P}_\theta(\mathbf{X} \in S_1)$, defined for all $\theta \in \Omega$.

(iv) **Size:** $\alpha(\delta) = \sup_{\theta \in \Omega_0} \pi(\theta \mid \delta)$. The test δ is *level* α_0 iff $\alpha(\delta) \le \alpha_0$, i.e. the worst-case Type I error rate over Ω_0 is bounded by α_0.

Relations:

$$\sup_{\theta \in \Omega_0} \pi(\theta \mid \delta) = \alpha(\delta) = \sup_{\theta \in \Omega_0} \mathbb{P}(\text{Type I}), \qquad \sup_{\theta \in \Omega_1} \big[1 - \pi(\theta \mid \delta)\big] = \sup_{\theta \in \Omega_1} \mathbb{P}(\text{Type II}).$$

Equivalently, $\inf_{\theta \in \Omega_1} \pi(\theta \mid \delta)$ is the worst-case power.

A2. (3 pts) Neyman–Pearson Lemma.

Let f_0 and f_1 be the joint densities under $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$. Suppose δ' rejects H_0 according to:

- Reject if $f_1(\mathbf{x}) > k\, f_0(\mathbf{x})$,
- Do not reject if $f_1(\mathbf{x}) < k\, f_0(\mathbf{x})$,
- Either decision allowed if $f_1(\mathbf{x}) = k\, f_0(\mathbf{x})$.

Then for any other test δ:

$$\alpha(\delta) \le \alpha(\delta') \implies \beta(\delta) \ge \beta(\delta'),$$

i.e. δ' minimizes the Type II error among all tests with at most the same Type I error.

When does NP yield a UMP test against a one-sided composite alternative? If the family has *monotone likelihood ratio (MLR)* in some statistic T, then the NP-optimal rejection

region $\{T > c\}$ does not depend on the specific value of $\theta_1 \in \Omega_1$; the same test is optimal against every θ_1 in the one-sided alternative, so it is UMP for $H_0 \colon \theta \le \theta_0$ vs. $H_1 \colon \theta > \theta_0$.

A3. (2 pts) Test/CI duality.

Given a $1-\alpha_0$ confidence set $\mathcal{S}(\mathbf{X})$, define the test of $H_0 \colon \theta = \theta_0$ versus $H_1 \colon \theta \neq \theta_0$ by:

$$\text{reject } H_0 \iff \theta_0 \notin \mathcal{S}(\mathbf{X}).$$

This test has level α_0 because, by the coverage property of the confidence set,

$$\mathbb{P}_{\theta_0}\big(\theta_0 \notin \mathcal{S}(\mathbf{X})\big) = 1 - \mathbb{P}_{\theta_0}\big(\theta_0 \in \mathcal{S}(\mathbf{X})\big) \le 1 - (1-\alpha_0) = \alpha_0.$$

A4. (2 pts) Wilks' theorem.

Under H_0, as $n \to \infty$,

$$-2\log\Lambda(\mathbf{X}) \xrightarrow{d} \chi^2_k,$$

where k is the difference in dimension between Ω and Ω_0 (i.e. the number of constraints H_0 imposes). **Conditions:** (i) the parameter space Ω is an open subset of $\mathbb{R}^p$ and Ω_0 is a $(p-k)$-dimensional smooth submanifold; (ii) the model satisfies the standard regularity conditions for asymptotic normality of the MLE (smoothness of $\log f$, finite Fisher information, support not depending on θ, ability to interchange differentiation and integration); (iii) the true $\theta \in \Omega_0$ lies in the interior of Ω_0.

Part B — Mid-length Testing Problems (30 points)

B1. One-sample t-test, p-value, and CI duality. (15 pts)

Given $n = 25$, $\overline{X} = 22.4$, $s^2 = 9.0$, so $s = 3$ and $s/\sqrt{n} = 3/5 = 0.6$.

(a) (5 pts) Test statistic

$$U_n = \frac{\overline{X} - \mu_0}{s/\sqrt{n}} = \frac{22.4 - 20}{0.6} = \frac{2.4}{0.6} = 4.0.$$

Under $H_0 \colon \mu = 20$, $U_n \sim t_{n-1} = t_{24}$. The two-sided level-0.05 rejection region is $|U_n| \ge t_{0.025,24} = 2.064$. Since $|4.0| = 4.0 > 2.064$, **reject** H_0.

(b) (3 pts) The two-sided p-value is $2[1 - T_{24}(|U_n|)] = 2[1 - T_{24}(4.0)]$. Since $|U_n| = 4.0$ is much larger than $t_{0.025,24} = 2.064$, we have $T_{24}(4.0) > T_{24}(2.064) = 0.975$, so

$$p\text{-value} = 2[1 - T_{24}(4.0)] < 2[1 - 0.975] = 0.05.$$

Therefore the p-value is **less than** 0.05 (in fact much smaller — $|U_n| = 4$ on 24 d.f. corresponds to a very small probability). This is consistent with rejection at level 0.05.

(c) (4 pts) The 95% CI for μ is

$$\overline{X} \pm t_{0.025,24} \cdot \tfrac{s}{\sqrt{n}} = 22.4 \pm 2.064 \cdot 0.6 = 22.4 \pm 1.2384 = (21.16,\ 23.64).$$

Since $\mu_0 = 20$ is *not* in this interval, we conclude that $\mu \neq 20$ at the 5% level — the same conclusion as in (a). This is exactly Theorem 10.3 (test/CI duality): the level-α_0 two-sided test of $H_0 \colon \mu = \mu_0$ rejects iff $\mu_0 \notin (\overline{X} - t_{\alpha_0/2} \cdot s/\sqrt{n},\ \overline{X} + t_{\alpha_0/2} \cdot s/\sqrt{n})$.

(d) (3 pts) The one-sided test of $H_0\colon \mu \le 20$ versus $H_1\colon \mu > 20$ at $\alpha_0 = 0.05$ rejects when

$$U_n \ge t_{0.05,24} = 1.711.$$

Since $U_n = 4.0 > 1.711$, **reject** H_0 at the 5% level. (Note: the threshold drops from 2.064 to 1.711 because the entire $\alpha_0 = 0.05$ is allocated to one tail.)

B2. Two-sample t-test and F-test. (15 pts)

Given $m = n = 8$, $\overline{X} = 14.0$, $\overline{Y} = 11.5$, $S_X^2 = 16.0$, $S_Y^2 = 9.0$.

(a) (5 pts) The F-statistic is

$$V_{m,n} = \frac{S_X^2/(m-1)}{S_Y^2/(n-1)} = \frac{16/7}{9/7} = \frac{16}{9} \approx 1.778.$$

Under $H_0\colon \sigma_X^2 = \sigma_Y^2$, $V_{m,n} \sim F_{7,7}$. At $\alpha_0 = 0.10$ two-sided we reject if

$$V \le F_{0.95,7,7} = 1/3.79 \approx 0.264 \quad \text{or} \quad V \ge F_{0.05,7,7} = 3.79.$$

Since $0.264 < 1.778 < 3.79$, **fail to reject** the equal-variance hypothesis. The data are consistent with $\sigma_X^2 = \sigma_Y^2$, so it is reasonable to proceed with the pooled-variance two-sample t-test.

(b) (5 pts) Pooled-variance two-sample t-statistic:

$$\begin{aligned}
\tfrac{1}{m} + \tfrac{1}{n} &= \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}, \\
S_X^2 + S_Y^2 &= 16 + 9 = 25, \\
\left(\tfrac{1}{m} + \tfrac{1}{n}\right)(S_X^2 + S_Y^2) &= \tfrac{1}{4} \cdot 25 = \tfrac{25}{4}, \\
\sqrt{\left(\tfrac{1}{m} + \tfrac{1}{n}\right)(S_X^2 + S_Y^2)} &= \tfrac{5}{2}, \\
\sqrt{m+n-2} &= \sqrt{14}, \\
U_{m,n} &= \frac{\sqrt{14}\,(14.0 - 11.5)}{5/2} = \frac{\sqrt{14} \cdot 2.5}{2.5} = \sqrt{14} \approx 3.742.
\end{aligned}$$

(c) (3 pts) Under $H_0\colon \mu_X = \mu_Y$, $U_{m,n} \sim t_{m+n-2} = t_{14}$. Reject if $|U_{m,n}| \ge t_{0.025,14} = 2.145$. Since $\sqrt{14} \approx 3.742 > 2.145$, **reject** H_0: there is significant evidence that $\mu_X \ne \mu_Y$ at the 5% level.

(d) (2 pts) The corresponding 95% CI for $\mu_X - \mu_Y$ (by inverting the level-0.05 acceptance region) is

$$(\overline{X} - \overline{Y}) \pm t_{0.025,14} \cdot \frac{\sqrt{(\frac{1}{m} + \frac{1}{n})(S_X^2 + S_Y^2)}}{\sqrt{m+n-2}} = 2.5 \pm 2.145 \cdot \frac{5/2}{\sqrt{14}} = 2.5 \pm 2.145 \cdot 0.668.$$

Numerically: $2.5 \pm 1.434 = (1.066,\ 3.934)$. The interval excludes 0, consistent with rejection in (c).

Part C — Major Problems (60 points)

C1. Neyman–Pearson and UMP for the exponential rate. (18 pts)

(a) (6 pts) The joint density is

$$f(\mathbf{x} \mid \lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda T}, \qquad T = \sum_{i=1}^{n} x_i.$$

For $\lambda_1 > \lambda_0 > 0$, the likelihood ratio is

$$\frac{f(\mathbf{x} \mid \lambda_1)}{f(\mathbf{x} \mid \lambda_0)} = \left(\frac{\lambda_1}{\lambda_0}\right)^n \exp\bigl[-(\lambda_1 - \lambda_0)\, T\bigr].$$

The NP lemma says reject H_0 when this ratio exceeds k:

$$\left(\frac{\lambda_1}{\lambda_0}\right)^n e^{-(\lambda_1-\lambda_0)T} > k \iff -(\lambda_1 - \lambda_0)\, T > \log k - n \log(\lambda_1/\lambda_0) \iff T < c,$$

where $c = \dfrac{n \log(\lambda_1/\lambda_0) - \log k}{\lambda_1 - \lambda_0}$.

The inequality flipped because $\lambda_1 - \lambda_0 > 0$, so dividing both sides reverses the sign. **Intuition:** a larger rate λ produces *smaller* observations on average, hence smaller T. Small T is evidence for the alternative $\lambda_1 > \lambda_0$.

Most powerful test: reject H_0 iff $T \le c$.

(b) (4 pts) Under H_0, $T = \sum_{i=1}^{n} X_i$ is a sum of n i.i.d. $\mathrm{Exp}(\lambda_0)$ random variables, so

$$T \sim \mathrm{Gamma}(n, \lambda_0)$$

in DeGroot's rate parameterization, with density $f_T(t) = \frac{\lambda_0^n}{(n-1)!} t^{n-1} e^{-\lambda_0 t}$ for $t > 0$.

To get size $\alpha_0 = 0.05$, choose c so that $\mathbb{P}_{\lambda_0}(T \le c) = 0.05$, i.e. $c = G^{-1}_{n,\lambda_0}(0.05)$, the 0.05-quantile of $\mathrm{Gamma}(n, \lambda_0)$. Equivalently, $2\lambda_0 T \sim \chi^2_{2n}$ under H_0, so

$$c = \frac{1}{2\lambda_0}\, \chi^2_{0.05,\, 2n} = \frac{1}{2\lambda_0} \cdot \chi^2_{0.05,40} \quad (\text{since } n = 20).$$

(c) (4 pts) The Exponential family with rate has *monotone likelihood ratio* in $-T$ (equivalently, decreasing MLR in T): for any $\lambda_1 > \lambda_0$, the ratio $f_{\mathbf{x}}(\lambda_1)/f_{\mathbf{x}}(\lambda_0)$ is a decreasing function of T. Therefore the most powerful test from (a), which rejects for $T \le c$, takes the same form for *every* $\lambda_1 > \lambda_0$ — the cutoff c does not depend on the particular λ_1.

To confirm size control over $\Omega_0 = \{\lambda \le \lambda_0\}$: for any $\lambda \le \lambda_0$,

$$\pi(\lambda) = \mathbb{P}_\lambda(T \le c).$$

Since T under rate λ is stochastically larger when λ is smaller (smaller rate $\Rightarrow$ longer waits $\Rightarrow$ larger T), $\mathbb{P}_\lambda(T \le c)$ is non-decreasing in λ. Hence $\sup_{\lambda \le \lambda_0} \pi(\lambda) = \pi(\lambda_0) = \alpha_0$. Combined with the previous paragraph, the test is UMP at level α_0.

(d) (4 pts) For any $\lambda > 0$,

$$\pi(\lambda) = \mathbb{P}_\lambda(T \le c) = \int_0^c \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}\, dt = G_{n,\lambda}(c),$$

where $G_{n,\lambda}$ is the cdf of Gamma(n, λ). Equivalently, since $2\lambda T \sim \chi^2_{2n}$,

$$\pi(\lambda) = \mathbb{P}\big(\chi^2_{2n} \le 2\lambda c\big) = F_{\chi^2_{2n}}(2\lambda c).$$

This is increasing in λ (the test has more power against larger λ, as expected).

C2. Likelihood ratio test for a normal variance. (18 pts)

(a) (8 pts) The likelihood is

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right).$$

Let $S = \sum_{i=1}^n (X_i - \overline{X})^2$.

Numerator (sup over Ω_0): $\sigma^2 = \sigma_0^2$ is fixed; the only free parameter is μ. The MLE of μ is $\overline{X}$, and at this value $\sum (X_i - \mu)^2 = S$. Hence

$$\sup_{\Omega_0} L = (2\pi\sigma_0^2)^{-n/2} \exp\left(-\frac{S}{2\sigma_0^2}\right).$$

Denominator (sup over Ω): both μ and σ^2 free; MLEs are $\widehat{\mu} = \overline{X}$ and $\widehat{\sigma}^2 = S/n$. At these values,

$$\sup_{\Omega} L = (2\pi\widehat{\sigma}^2)^{-n/2} \exp\left(-\frac{S}{2\widehat{\sigma}^2}\right) = (2\pi S/n)^{-n/2}\, e^{-n/2}.$$

Therefore

$$\Lambda_n(\mathbf{X}) = \frac{\sup_{\Omega_0} L}{\sup_{\Omega} L} = \frac{(2\pi\sigma_0^2)^{-n/2} \exp(-S/(2\sigma_0^2))}{(2\pi S/n)^{-n/2}\, e^{-n/2}} = \left(\frac{S/n}{\sigma_0^2}\right)^{n/2} \exp\left(\frac{n}{2} - \frac{S}{2\sigma_0^2}\right).$$

Define $T = S/\sigma_0^2 = \sum (X_i - \overline{X})^2/\sigma_0^2$. Then $S/(n\sigma_0^2) = T/n$, and

$$\boxed{\Lambda_n(\mathbf{X}) = \left(\frac{T}{n}\right)^{n/2} \exp\left(\frac{n-T}{2}\right).}$$

Λ_n is a function of T alone.

(b) (5 pts) Take logs:

$$\log \Lambda_n = \frac{n}{2} \log(T/n) + \frac{n-T}{2} = \frac{n}{2}\big[\log(T/n) + 1 - T/n\big].$$

Setting $u = T/n$, $g(u) = \log u + 1 - u$ has derivative $g'(u) = 1/u - 1$, so g is uniquely maximized at $u = 1$ (i.e. $T = n$) with $g(1) = 0$. $g(u) \to -\infty$ as $u \to 0^+$ or $u \to \infty$. Therefore $\log \Lambda_n$ (and hence Λ_n) is a single-peaked function of T with maximum $\Lambda_n = 1$ at $T = n$, decreasing on both sides.

It follows that for any threshold $k \in (0,1)$, $\{\Lambda_n \le k\}$ is equivalent to $\{T \le c_1\} \cup \{T \ge c_2\}$ for constants $c_1 < n < c_2$ chosen so that the size equals α_0.

Under $H_0 \colon \sigma^2 = \sigma_0^2$, the standard result gives

$$T = \frac{\sum (X_i - \overline{X})^2}{\sigma_0^2} \sim \chi^2_{n-1}.$$

The standard equal-tailed χ^2-test takes $c_1 = \chi^2_{\alpha_0/2,\, n-1}$ and $c_2 = \chi^2_{1-\alpha_0/2,\, n-1}$. This isn't exactly equal-tailed in the LRT sense (the LRT would split based on equal Λ_n values rather than equal tail probabilities) but it is the standard chi-square test and is asymptotically equivalent.

(c) (5 pts) With $n = 9$, $\sigma_0^2 = 4$, observed $\sum (X_i - \overline{X})^2 = 32$:

$$T = \frac{32}{4} = 8.$$

Under H_0, $T \sim \chi^2_8$. The level-0.05 equal-tailed rejection region is

$$\{T \le \chi^2_{0.025,8}\} \cup \{T \ge \chi^2_{0.975,8}\} = \{T \le 2.18\} \cup \{T \ge 17.53\}.$$

Since $2.18 < 8 < 17.53$, T falls in the acceptance region. **Fail to reject** H_0: the data are consistent with $\sigma^2 = 4$. (Note that $E[T] = n - 1 = 8$ under H_0, and we observed exactly $T = 8$, so this is the most consistent possible value.)

C3. Support-dependent family — direct Type I/II calculation. (12 pts)

The pdf $f(x \mid \theta) = 2(1-\theta)x + \theta$ on $[0,1]$ has:

- $\int_0^1 [2(1-\theta)x + \theta]\, dx = (1-\theta) + \theta = 1$ (valid pdf), and
- $f(x \mid \theta) \ge 0$ for $\theta \in [0,1]$ (since both terms are non-negative).

The test rejects $H_0 \colon \theta \ge 1/2$ when $X > 1/3$, so the power function is

$$\pi(\theta) = \mathbb{P}_\theta(X > 1/3) = \int_{1/3}^1 \big[2(1-\theta)x + \theta\big]\, dx.$$

Compute:

$$\int_{1/3}^1 2x\, dx = \big[x^2\big]_{1/3}^1 = 1 - \tfrac{1}{9} = \tfrac{8}{9},$$
$$\int_{1/3}^1 1\, dx = 1 - \tfrac{1}{3} = \tfrac{2}{3}.$$

Therefore

$$\pi(\theta) = (1-\theta) \cdot \tfrac{8}{9} + \theta \cdot \tfrac{2}{3} = \tfrac{8}{9} - \tfrac{8}{9}\theta + \tfrac{6}{9}\theta = \tfrac{8}{9} - \tfrac{2}{9}\theta.$$

(a) (5 pts) Type I error. $\pi(\theta) = \frac{8}{9} - \frac{2}{9}\theta$ is *decreasing* in θ (slope $-2/9 < 0$). Hence on $\Omega_0 = [1/2, 1]$ the supremum is attained at the smallest θ, namely $\theta = 1/2$:

$$\alpha(\delta) = \sup_{\theta \ge 1/2} \pi(\theta) = \pi(1/2) = \tfrac{8}{9} - \tfrac{2}{9} \cdot \tfrac{1}{2} = \tfrac{8}{9} - \tfrac{1}{9} = \tfrac{7}{9}.$$

This is a *very* large Type I error — the test rejects too readily because the cutoff $X > 1/3$ excludes only a small portion of the unit interval. The rejection region should be moved closer to 1 to reduce α.

(b) (3 pts) Type II error at $\theta = 0$.

$$1 - \pi(0) = 1 - \tfrac{8}{9} = \tfrac{1}{9}.$$

(c) (2 pts) Cramér–Rao does not apply (or is irrelevant) here for two reasons:

- The parameter space $\Theta = [0, 1]$ is *closed*, so the regularity condition that θ lie in the interior of an open set fails at the endpoints. Standard differentiation-under-the-integral arguments break at $\theta \in \{0, 1\}$.
- More fundamentally, the CR lower bound is a lower bound on the variance of *unbiased estimators*; here we are computing exact Type I/II errors of a fixed test, not assessing an estimator's efficiency. The CR bound is silent about the power function of a test.

(Note: the support $[0, 1]$ does *not* depend on θ, so the classical "support depends on θ" failure does not apply.)

(d) (2 pts) Sketch of $\pi(\theta)$. $\pi(\theta) = \frac{8}{9} - \frac{2}{9}\theta$ is a straight line decreasing from $\pi(0) = 8/9$ to $\pi(1) = 6/9 = 2/3$. Specifically:

$$\pi(0) = \tfrac{8}{9} \approx 0.889, \qquad \pi(1/2) = \tfrac{7}{9} \approx 0.778, \qquad \pi(1) = \tfrac{6}{9} = \tfrac{2}{3} \approx 0.667.$$

θ	0	1/2	1
$\pi(\theta)$	8/9	7/9	6/9

The line slopes gently downward over $[0, 1]$. A "good" test would have π small on $\Omega_0 = [1/2, 1]$ and large on $\Omega_1 = [0, 1/2)$; here π is roughly the same value across the entire parameter space, indicating that this test does poorly at distinguishing H_0 from H_1.

C4. Cumulative review — sufficiency, Bayes, consistency. (12 pts)

(a) (4 pts) The joint pmf is

$$f(\mathbf{x} \mid p) = \prod_{i=1}^{n} p(1-p)^{x_i} = p^n (1-p)^{\sum x_i} = p^n (1-p)^T, \quad T = \sum x_i.$$

Define $g(T, p) = p^n(1-p)^T$ and $h(\mathbf{x}) = 1$ (subject to $x_i \in \{0, 1, 2, \dots\}$, which we encode as an indicator). Then

$$f(\mathbf{x} \mid p) = g(T(\mathbf{x}), p) \cdot h(\mathbf{x}).$$

By the Fisher–Neyman factorization criterion, $T = \sum_{i=1}^{n} X_i$ is a sufficient statistic for p.

(b) (4 pts) The prior density is $\xi(p) \propto p^{\alpha_0 - 1}(1-p)^{\beta_0 - 1}$ on $(0, 1)$. The posterior is proportional to prior $\times$ likelihood:

$$\xi(p \mid \mathbf{x}) \propto p^{\alpha_0 - 1}(1-p)^{\beta_0 - 1} \cdot p^n (1-p)^T = p^{(n+\alpha_0)-1}(1-p)^{(T+\beta_0)-1}.$$

This is the kernel of a Beta distribution:

$$p \mid \mathbf{X} = \mathbf{x} \;\sim\; \text{Beta}(n + \alpha_0,\; T + \beta_0).$$

(c) (2 pts) Under squared-error loss, the Bayes estimator equals the posterior mean. For Beta(a, b) the mean is $a/(a+b)$, so

$$\widehat{p}_{\text{Bayes}} = \mathbb{E}[p \mid \mathbf{X}] = \frac{n+\alpha_0}{(n+\alpha_0)+(T+\beta_0)} = \frac{n+\alpha_0}{n+\alpha_0+\beta_0+T}.$$

(d) (2 pts) For Geometric(p), $\mathbb{E}[X_i] = (1-p)/p$, so by the WLLN

$$\frac{T}{n} = \overline{X}_n \xrightarrow{P} \frac{1-p}{p}.$$

Then

$$\widehat{p}_{\text{MLE}} = \frac{n}{n+T} = \frac{1}{1+T/n} \xrightarrow{P} \frac{1}{1+(1-p)/p} = p.$$

For the Bayes estimator, divide numerator and denominator by n:

$$\widehat{p}_{\text{Bayes}} = \frac{1+\alpha_0/n}{1+(\alpha_0+\beta_0)/n+T/n} \xrightarrow{P} \frac{1+0}{1+0+(1-p)/p} = \frac{p}{p+(1-p)} = p.$$

By the continuous mapping theorem (or just Slutsky), both estimators converge in probability to p. The prior parameters α_0, β_0 are washed out as $n \to \infty$, which is the usual phenomenon: the data dominate the prior in the large-sample limit.